International Journal of Advanced Science and Technology
Vol.121 (2018), pp.43-54
http://dx.doi.org/10.14257/ijast.2018.121.04

Review of Different Image Data Hiding Techniques

Sonali Mishra[1], Ananya Dastidar[2*], Umesh Chandra Samal[3] and Sushanta Kumar Mohapatra[4]

[1,2]Department of Instrumentation and Electronics, College of Engineering and Technology, Bhubaneswar, Odisha, India
[3,4]School of Electronics Engineering, Kalinga Institute of Technology, Bhubaneswar, Odisha, India
[1]sonalimishra135@gmail.com [2]adastidar@cet.edu.in
[3]umesh.samalfetr@kiit.ac.in [4]skmctc74@gmail.com

Abstract

With the rapid progress of diverse communication technologies and the prevalence of computer and network technology, it is a big challenge to ensure secure transmission of information from source to destination. The World Wide Web is used as a medium to circulate a huge amount of digital information throughout the world every day. Most of these data contains valuable information that face the threat of being exposed and being corrupted. Encryption methods are evolving which is moving towards a future of limitless possibilities. Cryptography denotes the storing and transmission of information in a specific way so that the owner of the key only can access and process it further. This paper presents the study of different encryption and decryption techniques used for storing and transmitting of data and also implements the watermarking techniques and cryptography for study of encryption. We also propose a hybrid encryption technique based on watermarking and cryptography-the Secure Force Algorithm. The proposed hybrid technique gives us a promising result in terms of its mean square error and peak signal to noise ratio.

Keywords: *Cryptograph; Decryption techniques; Encryption techniques; Steganography; Watermarking*

1. Introduction

Today Security of Data has gained a lot of importance for all types of communication system. Extensive use of images, the text had made it more significant to focus on the area of data security. While the relentless use of Internet and Communication Technology is unavoidable, in the use of data sharing, many ways to provide security to data, are also being developed to look into the issues related to hacking, unethical sharing, etc. This paper reviews different literature based on various techniques used for data security in the field of communication. Image encryption is a method that sends out the image securely over various data communication networks to prevent the access of the secret information from illegal users. Some of the features of the image are its size, redundant nature and pixel correlation, which make it distinct from other data. Encryption techniques are well considered and very beneficial tool to protect the secret and valuable information from unauthorized and illegal users. Encryption converts the plain message to ciphertext which cannot be accessed by any unauthorized users. Decryption converts encrypted data to its original data form to make it accessible to the users. These techniques include encryption

Received (September 2, 2018), Review Result (November 3, 2018), Accepted (November 9, 2018)
*Corresponding Author

ISSN: 2005-4238 IJAST
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